

Lingesh R

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PROFILE

AI Engineer specializing in building real-world machine learning systems that integrate IoT sensors, signal processing, and deep learning. Experienced in developing end-to-end ML pipelines including data preprocessing, model training, and real-time inference using Python and Flask to deploy intelligent AI applications.

EDUCATION

B.Tech. Electronics and Communication Engineering

Sri Manakula Vinayagar Engineering College

CGPA: 8.5

2023 – 2027

Puducherry

AWARDS

Second Place in Hackathon 2026(State Level)

Meenakshi Engineering College

Best Real World Applicability Award- National Level Ideathon

Aarupadai Veedu Institute of Technology

SKILLS

- **Programming:** Python, Java, C
- **Machine Learning & Deep Learning:** Scikit-Learn, TensorFlow, Pytorch, Keras, OpenCV
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Natural Language Processing & LLMs:** LangChain, LangGraph, CrewAI, spaCy
- **Web Development:** Flask, FastAPI, HTML, CSS, Bootstrap
- **APIs, Deployment & Automation:** REST APIs, Streamlit, N8n, Git, GitHub, Postman
- **Web Scraping:** Selenium, BeautifulSoup
- **Databases & Tools:** MySQL, Jupyter Notebook, Google Colab, VS Code

PROJECTS

AI-Enabled Real-Time Sleep Apnea Detection Device | Python, TensorFlow, Flask, IoT, ESP32 

- Built an end-to-end AI + IoT system using ESP32 for real-time sleep apnea detection from breathing audio and vital signals.
- Extracted MFCC features and trained MLP on 103,210 breathing samples, achieving **81%** classification accuracy.
- Developed a Flask-based backend to ingest sensor data (INMP441 microphone, MAX30102 HR and SpO₂) and perform real-time inference.
- Integrated Twilio-based alerting with dual logic for apnea events and abnormal vital threshold detection.

Multi-Crop Disease Classification System | Python, TensorFlow, Flask, OpenCV 

- Built a Flask-based web application enabling users to select crop type (potato or tomato) and capture or upload leaf images for disease classification.
- Trained separate MLP models for potato and tomato leaf disease detection using 2,152 potato leaf images and 5,172 tomato leaf images.
- Applied data augmentation techniques to improve model generalization and robustness.
- Achieved **96.09%** accuracy for potato disease classification and **99.81%** accuracy for tomato disease classification.

Bengaluru Real Estate Price Prediction Engine | Python, Scikit-Learn, Flask, Feature Engineering 

- Developed a Flask-based web application with form-based user input to predict Bengaluru house prices.
- Engineered a backend inference pipeline to process features such as location, total_sqft, BHK, and bathrooms.
- Performed data preprocessing including missing value handling, outlier removal, and categorical feature encoding.
- Trained regression models achieving **85.9%** prediction accuracy for reliable price estimation.